**Cairo University**

**Faculty of Computers and Artificial Intelligence**

# CS251

# Introduction to Software Engineering

## $penderman

## Software Design Specifications

Version 1.0

Apr-2026

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Project: $penderman

# Software Design Specification

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# Software Design Specification

## Team

| ID | Name | Email | Mobile |
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## Document Purpose and Audience

**Document Purpose and Audience**

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**Document Purpose**

This Software Design Specification (SDS) serves as a comprehensive **technical blueprint** for the **$penderman** personal budgeting application. It translates functional requirements into a structured architectural design to guide the development and testing phases.

**Key objectives of this document include:**

- **Defining System Scope:** Outlining the boundaries of the application and its primary interactions with the "System User".
- **Structural Architecture:** Detailing the high-level technical containers, including the **JavaFX GUI**, the **Java Back-end**, and the **Microsoft SQL Server** database.
- **Behavioral Logic:** Providing step-by-step logic for critical operations such as user registration, transaction processing, and financial goal creation.
- **Data Integrity:** Establishing how financial data is persisted and retrieved through secure communication between the back-end and database layers.

# Software Design Specification

This document is intended for stakeholders involved in the technical execution and management of the project:

| Audience | Responsibility |
|---|---|
| **Software Developers** | To implement the specific classes, methods, and logic flows defined in the sequence diagrams. |
| **Project Managers** | To verify that the technical design aligns with the project's scope and business goals. |
| **Software Architects** | To ensure the structural integrity of the container-level interactions and technology stack. |
| **Quality Assurance (QA)** | To design test cases based on the documented logic for transactions and user management. |
| **Database Administrators** | To manage the SQL Server schema required to support the application's data persistence. |

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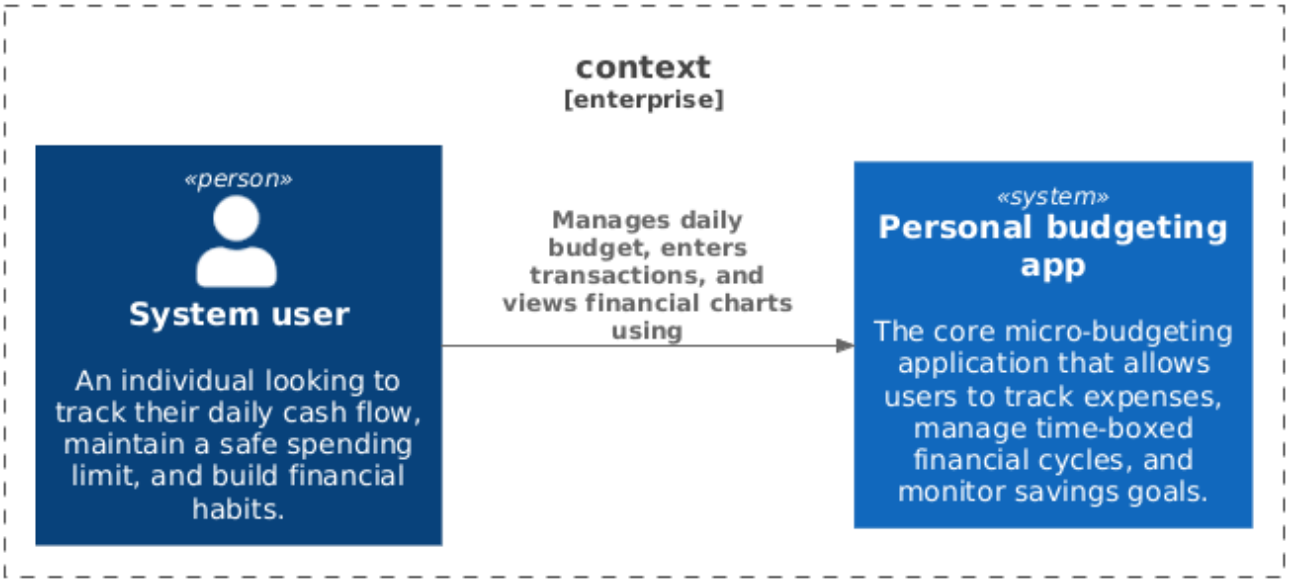
## System Models

### Architecture Diagram

#### 1.Context Diagram



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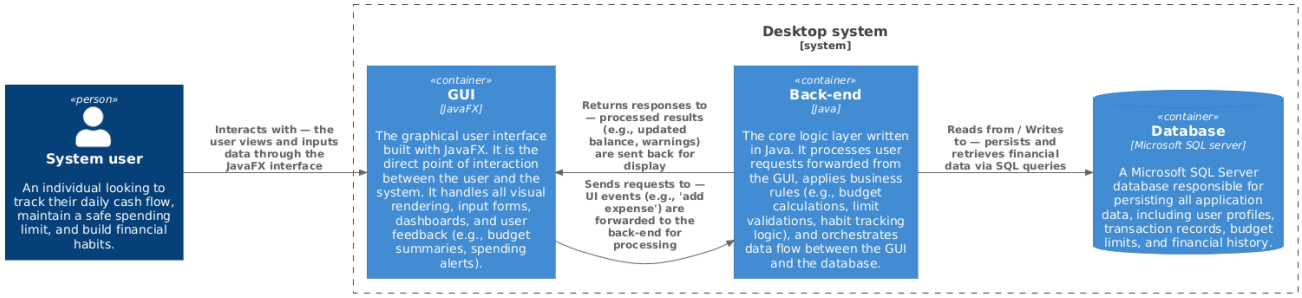
## 2.Container Diagram

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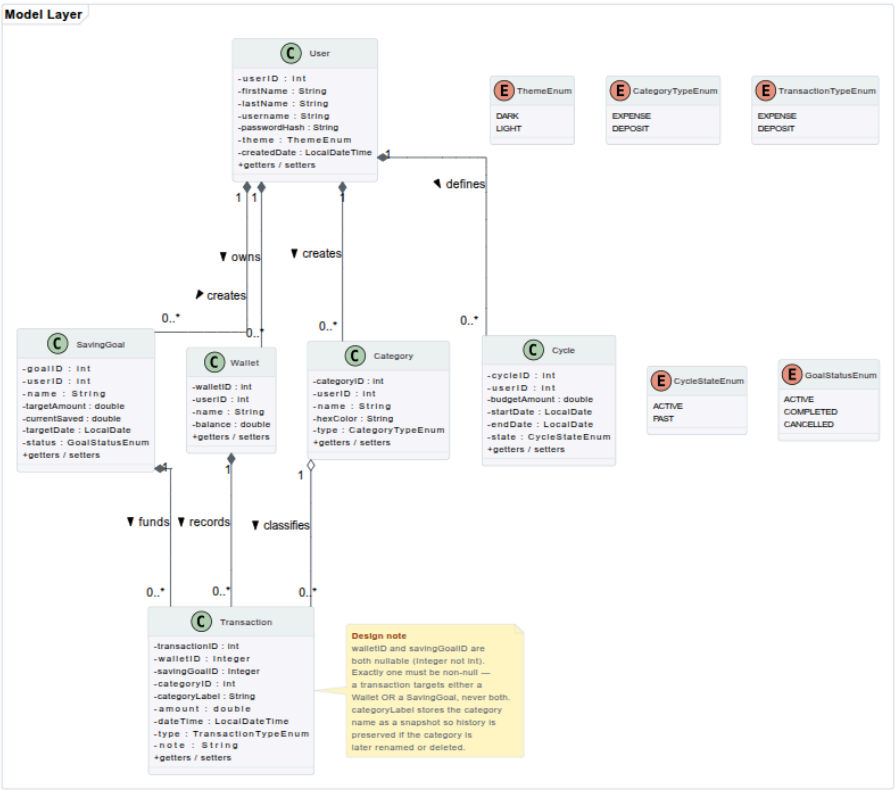
## Class Diagram(s)



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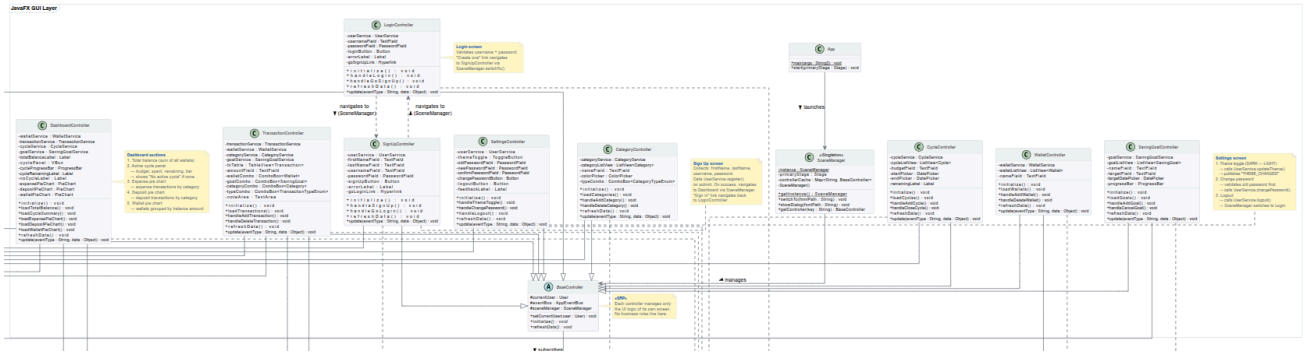

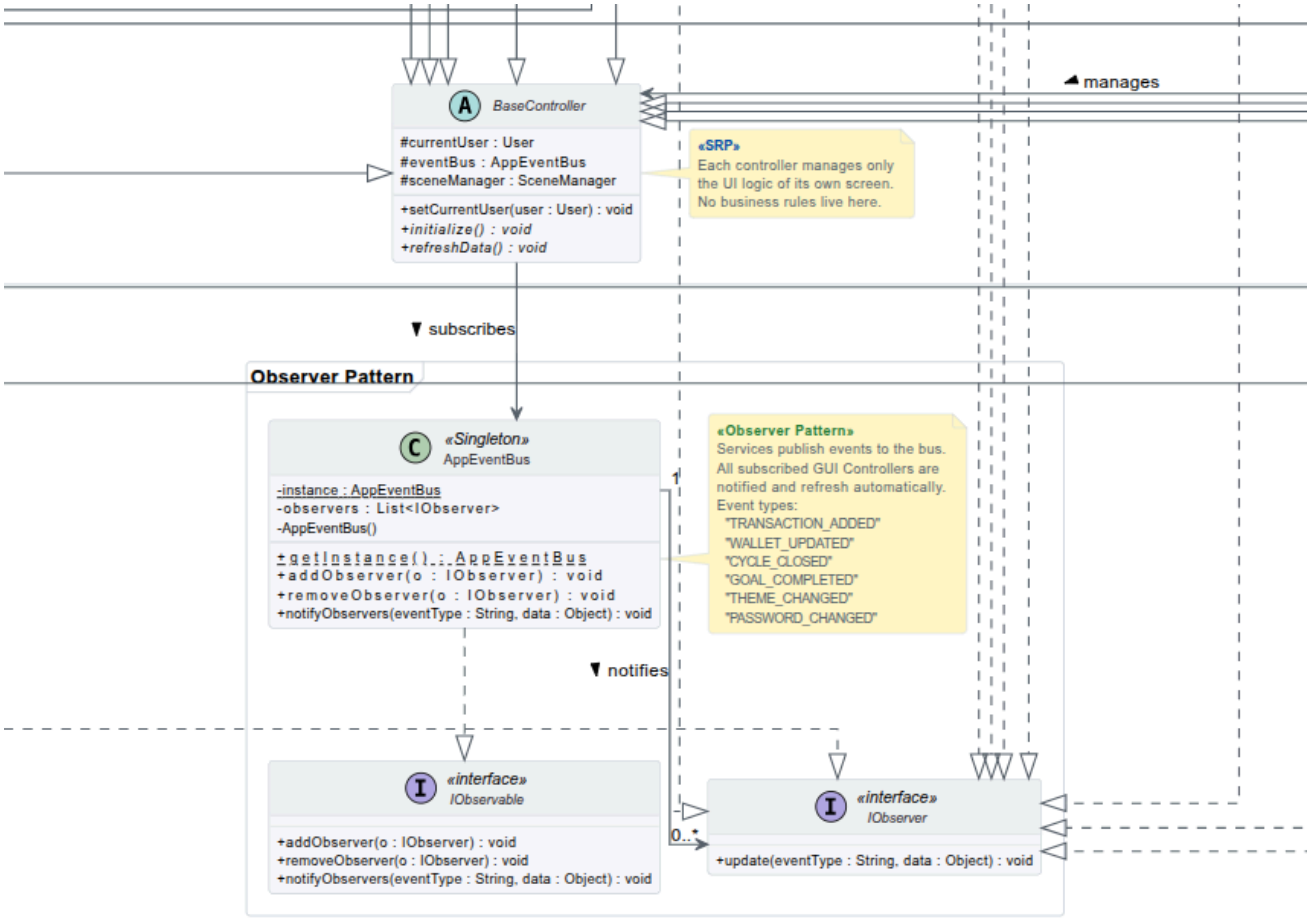

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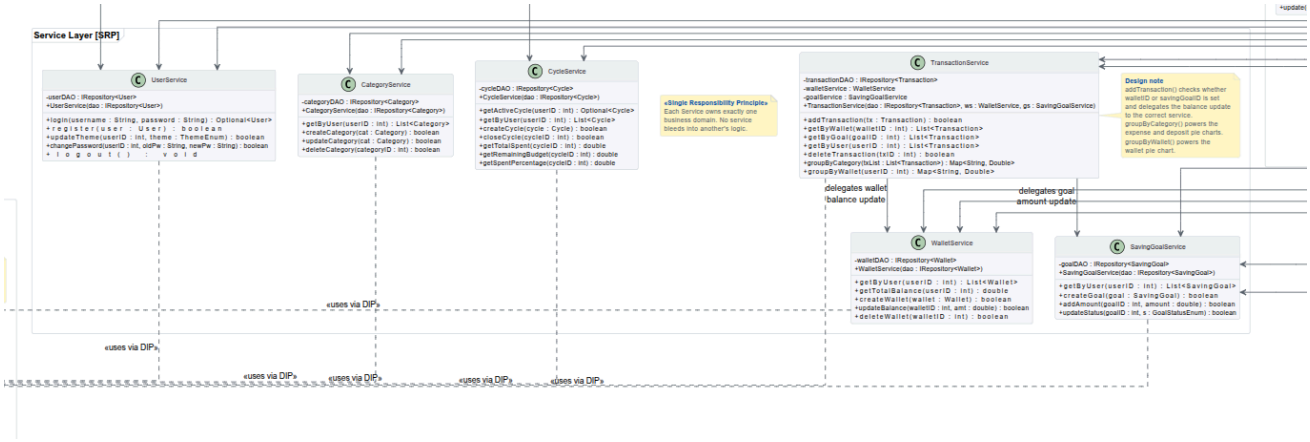


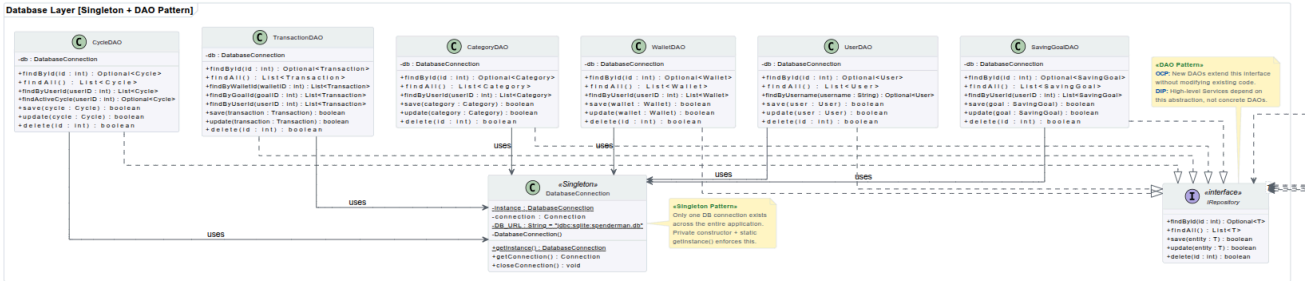


## For More details and high quality

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# Software Design Specification

## Class Descriptions

| Class Name | Description |
|---|---|
| **User** | The root owner of all system entities. Stores authentication details (username, hashed password) and user preferences (UI theme). |
| **Wallet** | Represents a financial account (e.g., "Cash"). Maintains a balance derived entirely from its recorded transactions. |
| **Category** | A user-defined label (e.g., "Food") with an assigned hex color for chart rendering. Distinguishes between expenses and deposits. |
| **Transaction** | The immutable ledger entry. Enforces mutual exclusivity: it must target exactly one Wallet OR one SavingGoal. Stores a static snapshot of the category name. |
| **Cycle** | A time-boxed budget period. Used by the system to calculate the safe daily spending limit over a specific date range. |
| **SavingGoal** | A financial milestone tracking saved amounts against a target. Automatically completes when fully funded. |
| **ThemeEnum** | Defines DARK or LIGHT UI preferences for persistent styling. |
| **CategoryTypeEnum** | Classifies categories as EXPENSE or DEPOSIT for UI filtering. |
| **TransactionTypeEnum** | Determines if a transaction adds (DEPOSIT) or subtracts (EXPENSE) from a balance. |
| **CycleStateEnum** | Tracks if a cycle is ACTIVE or PAST, ensuring only one active budget exists per user. |

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|---|---|
| **GoalStatusEnum** | Tracks milestone progress states: ACTIVE, COMPLETED, or CANCELLED. |
| **DatabaseConnection** | Provides a single, shared JDBC database connection globally, preventing performance overhead from multiple open instances. |
| **IRepository<T>** | Generic interface defining standard CRUD operations (findById, save, delete, etc.) to enable the Dependency Inversion Principle. |
| **UserDAO** | Handles user SQL queries, specifically managing findByUsername() for secure authentication workflows. |
| **WalletDAO** | Handles wallet SQL queries, executing frequent update() commands to adjust balances when transactions occur. |
| **CategoryDAO** | Handles category SQL queries, mapping UI hex colors and Enum types to the database schema. |
| **TransactionDAO** | Handles complex transaction queries. Explicitly manages the nullable walletID and savingGoalID foreign keys. |
| **CycleDAO** | Handles cycle SQL queries, utilizing findActiveCycle() as the foundation for the daily limit engine. |
| **SavingGoalDAO** | Handles saving goal SQL queries, updating progress amounts and lifecycle statuses. |
| **UserService** | Manages authentication, secure password hashing, registration, and theme updates. |
| **WalletService** | Manages wallet lifecycles and provides dynamic balance calculation across all of a user's accounts. |

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|---|---|
| **TransactionService** | The core logic hub. Enforces the Wallet XOR Goal rule, saves ledgers, and delegates balance updates to other services. |
| **CategoryService** | Safely manages category creation and deletion without corrupting historical ledger data. |
| **CycleService** | Calculates budget consumption and outputs the Safe Daily Limit engine data. Auto-archives past cycles. |
| **SavingGoalService** | Tracks goal contributions, triggering auto-completion when targets are met. |
| **IObserver** | The subscriber contract forcing all UI controllers to implement an update() method to listen for backend changes. |
| **IObservable** | The publisher contract defining methods to add, remove, and notify observers. |
| **AppEventBus** | The central radio tower (Pub/Sub hub). Services broadcast events here; Controllers listen here to automatically refresh their UI. |
| **App** | The application entry point. Bootstraps the JavaFX stage and hands control to the SceneManager. |
| **SceneManager** | The central navigation router. Caches built screens in memory to provide instantaneous page switching. |
| **BaseController** | The parent to all screens. Automatically connects child controllers to the EventBus and provides the logged-in user context. |
| **LoginController** | Authenticates users and passes the active session down to the application upon success. |

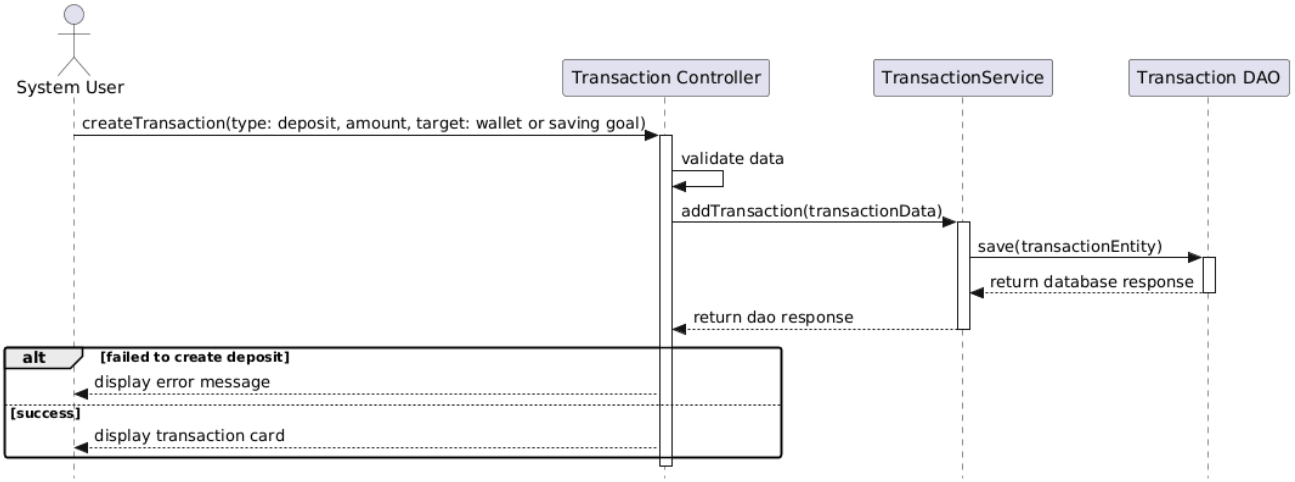
# Software Design Specification

| | |
|---|---|
| **SignUpController** | Handles new user registration with client-side validation and automatic post-creation login. |
| **DashboardController** | The primary data aggregator. Reactively updates pie charts, total balances, and the active cycle progress bar via events. |
| **TransactionController** | Manages the interactive ledger table. Features the dynamic Wallet/Goal XOR input form for strict data entry. |
| **WalletController** | Displays account balances and handles the creation of new financial containers. |
| **CategoryController** | Provides a visual color picker and type selector for generating new custom categories. |
| **CycleController** | Manages active budget periods and displays the computed Safe Daily Limit to the user. |
| **SavingGoalController** | Displays visual progress bars for financial targets. (Funding happens exclusively via the Transaction screen). |
| **SettingsController** | Handles global UI theme toggling, secure password changes, and safe session logout protocols. |

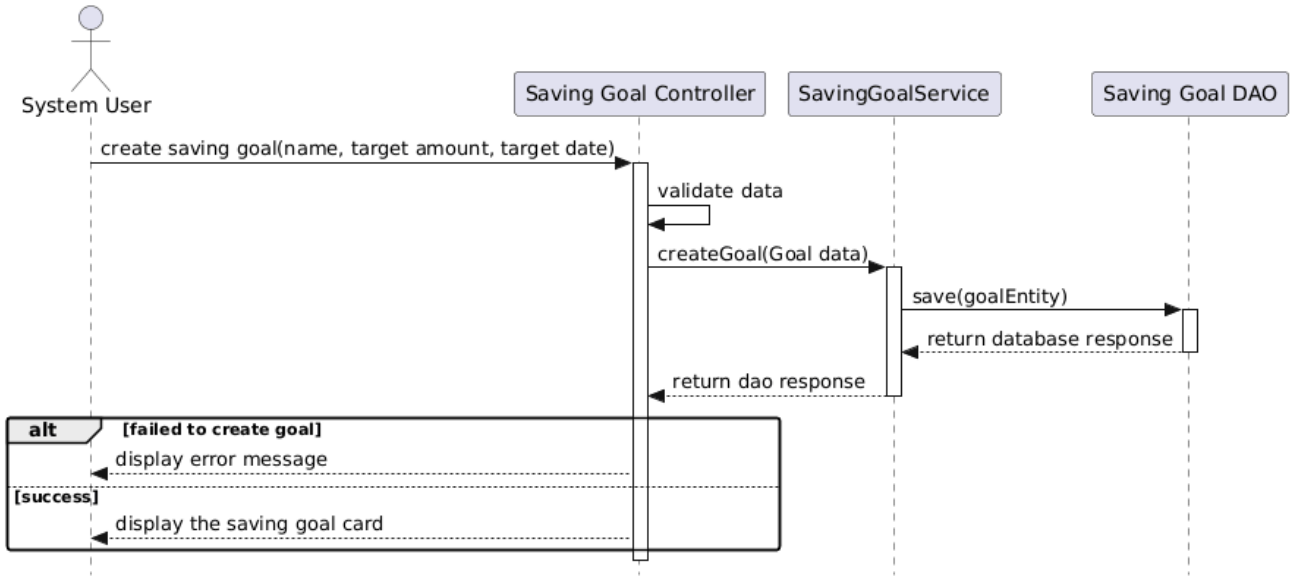
# Software Design Specification

## Sequence diagrams

### 1.Recording a Deposit

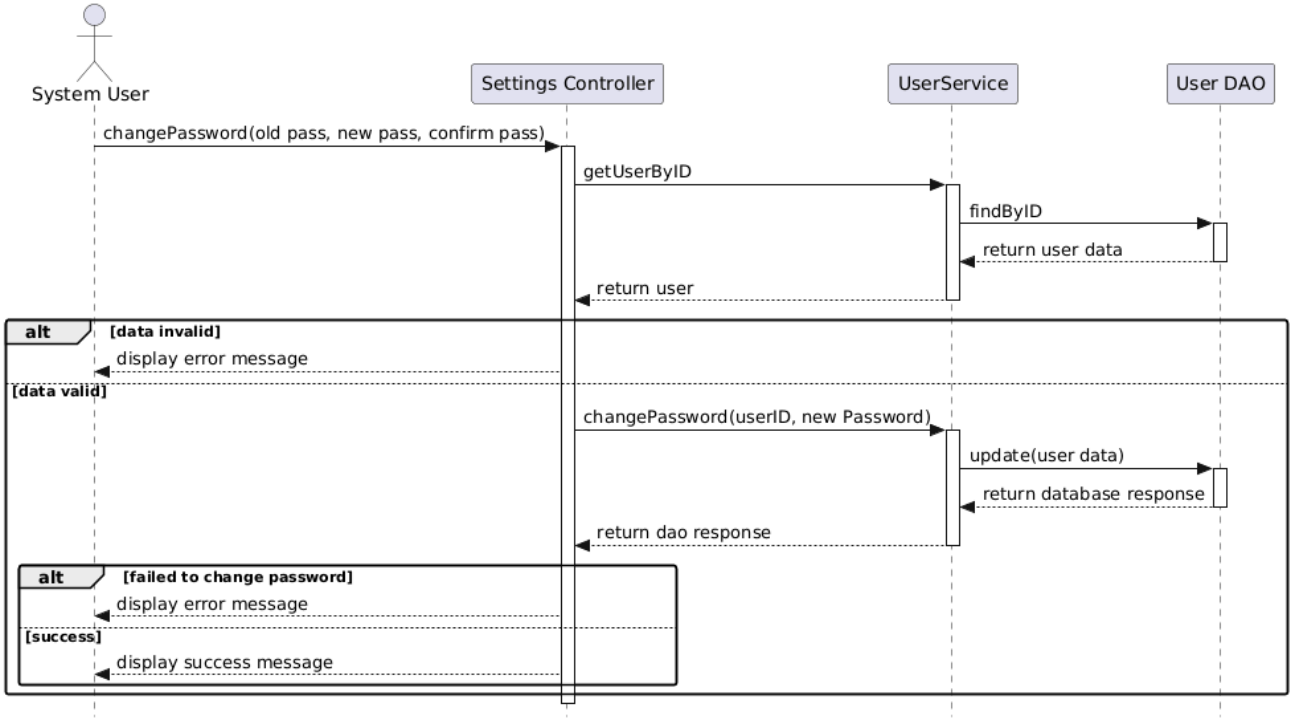



### 2.Creating Saving Goal



# Software Design Specification

## 3.Change Password



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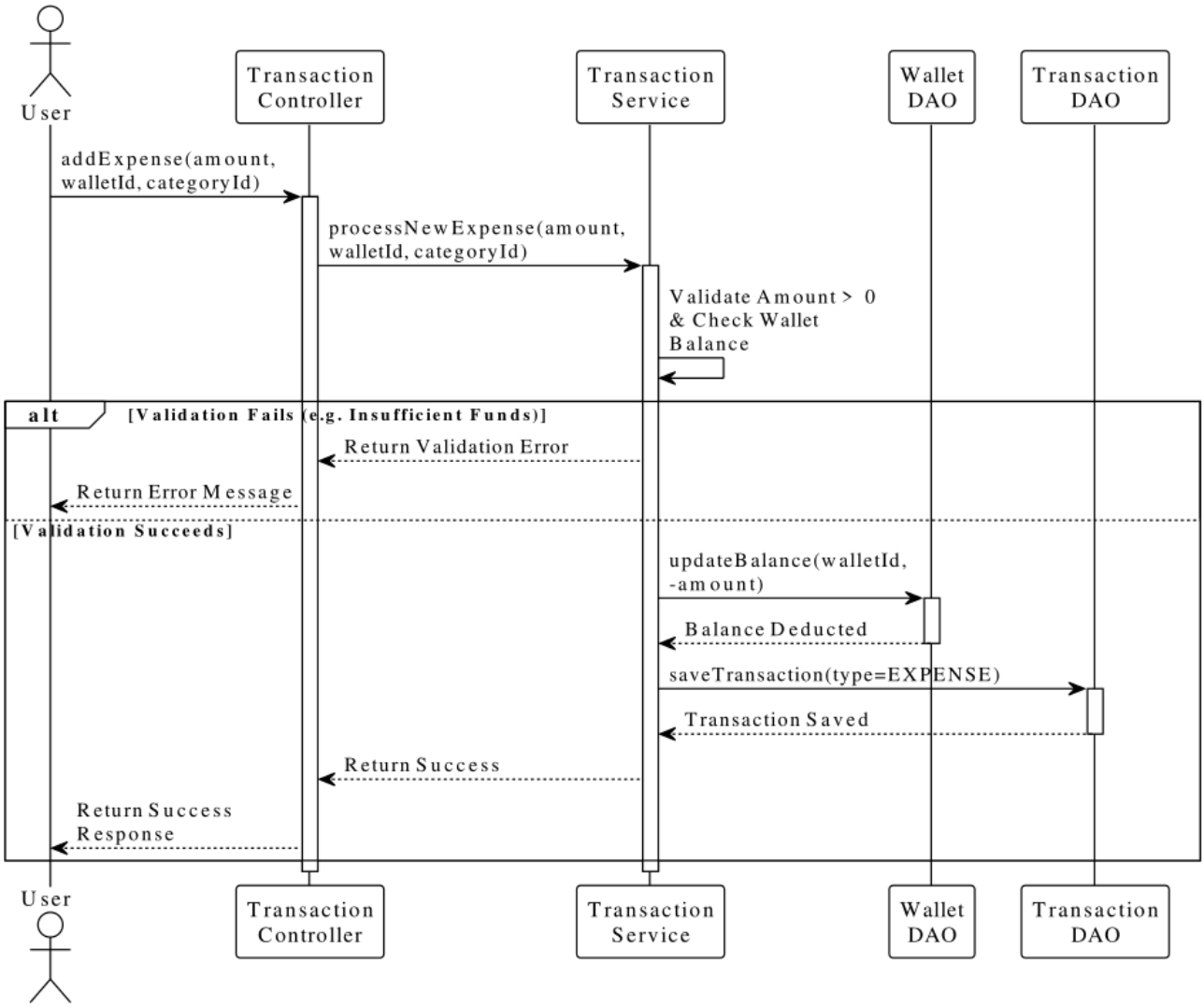
## 4.Logging Expense



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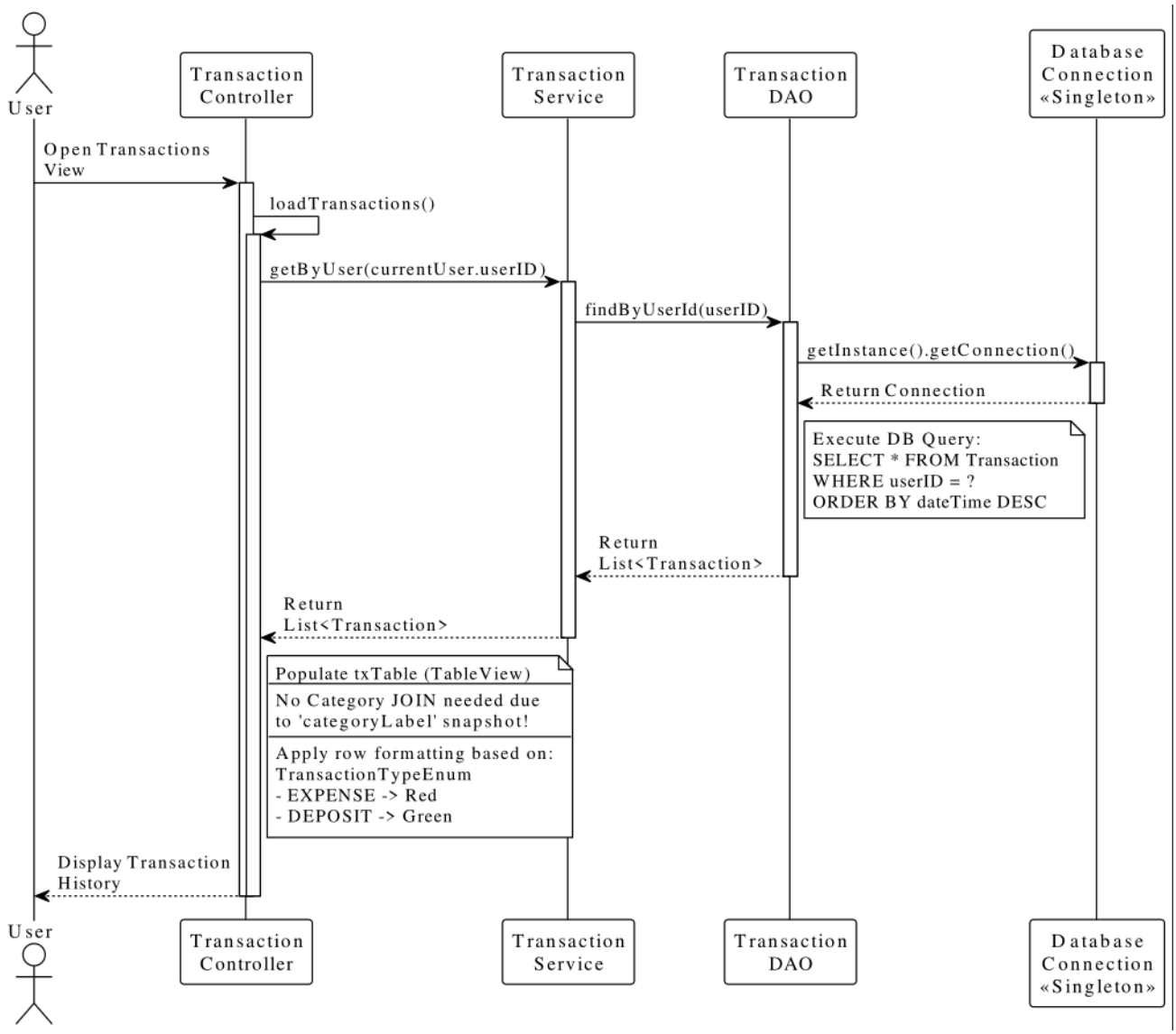
## 5.Viewing the Financial Ledger



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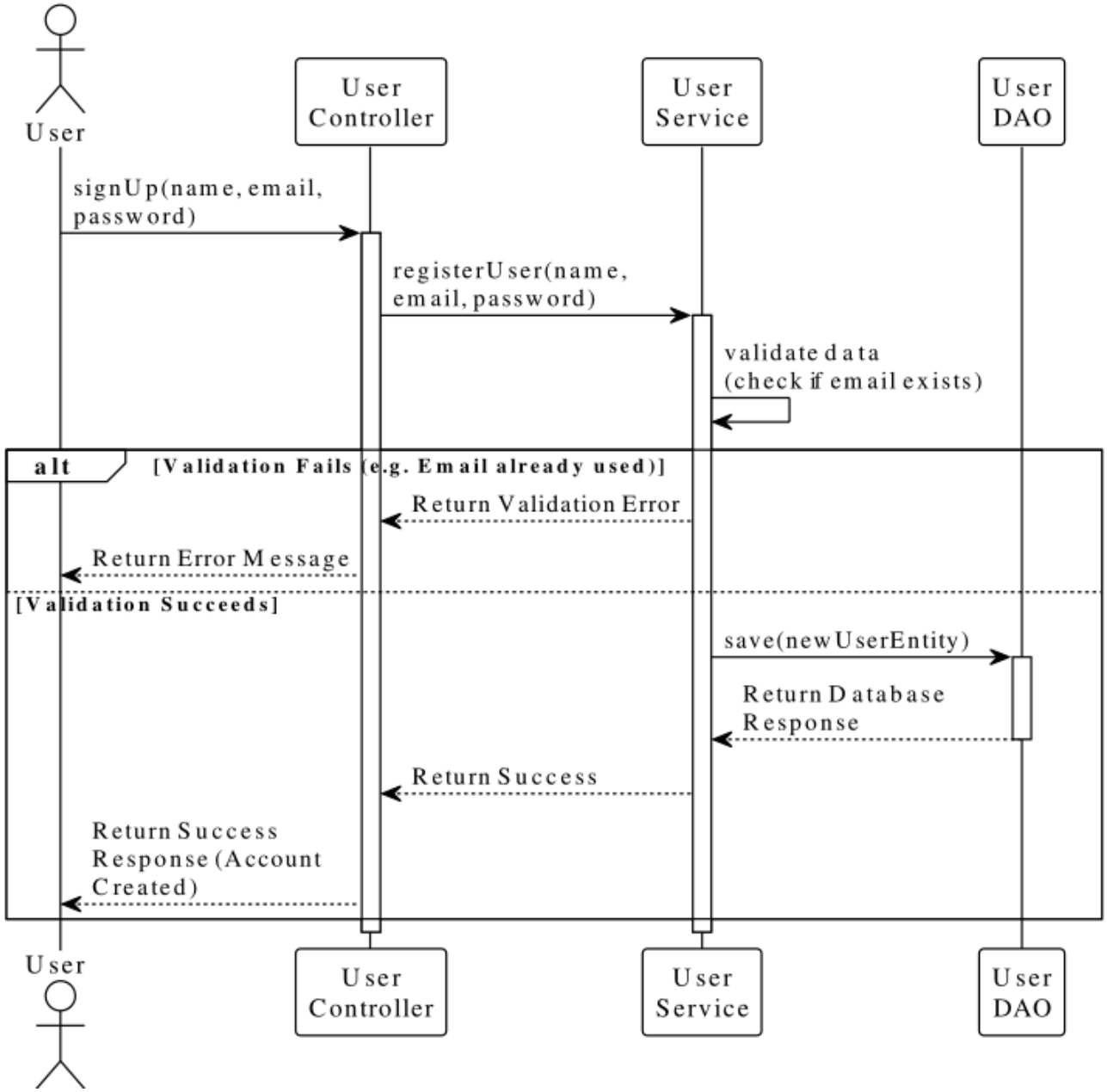
### 6.Sign Up



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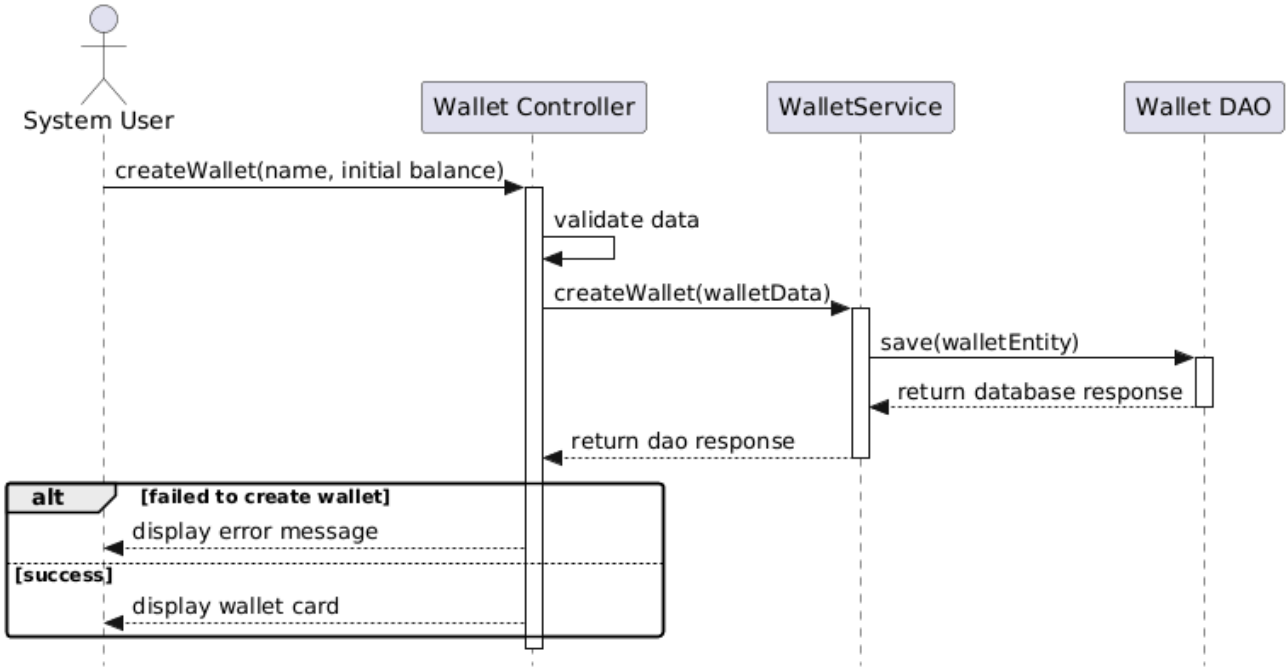
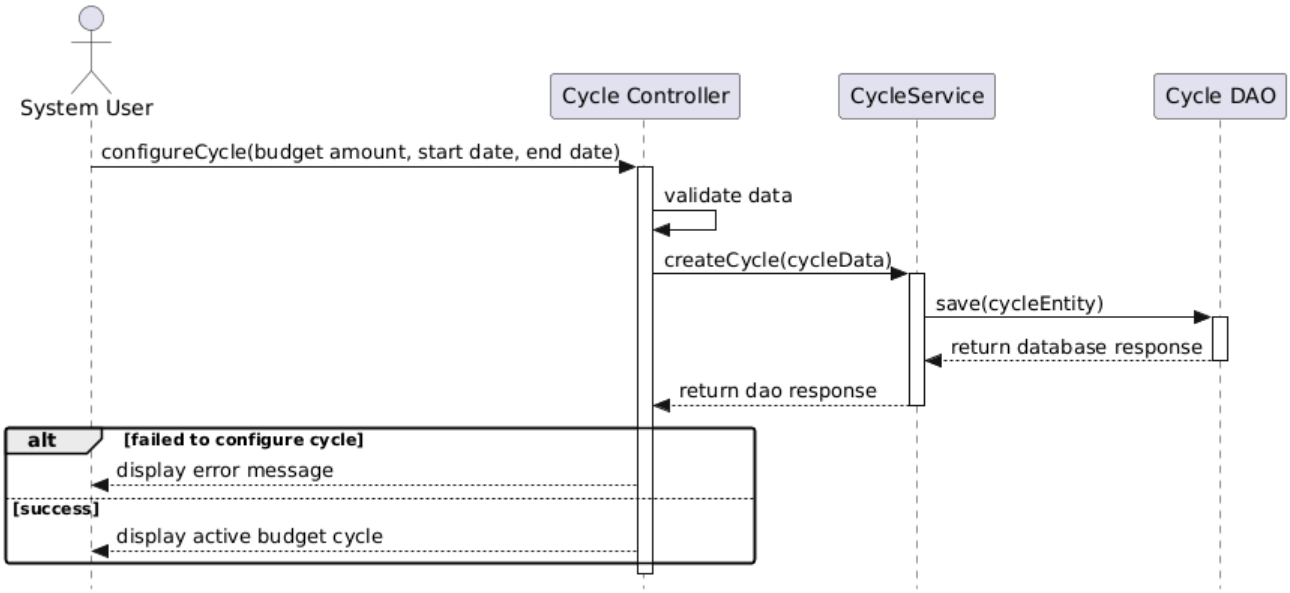
## 7.Custom Wallet Creation




## 8.Configuring an Active Budget Cycle



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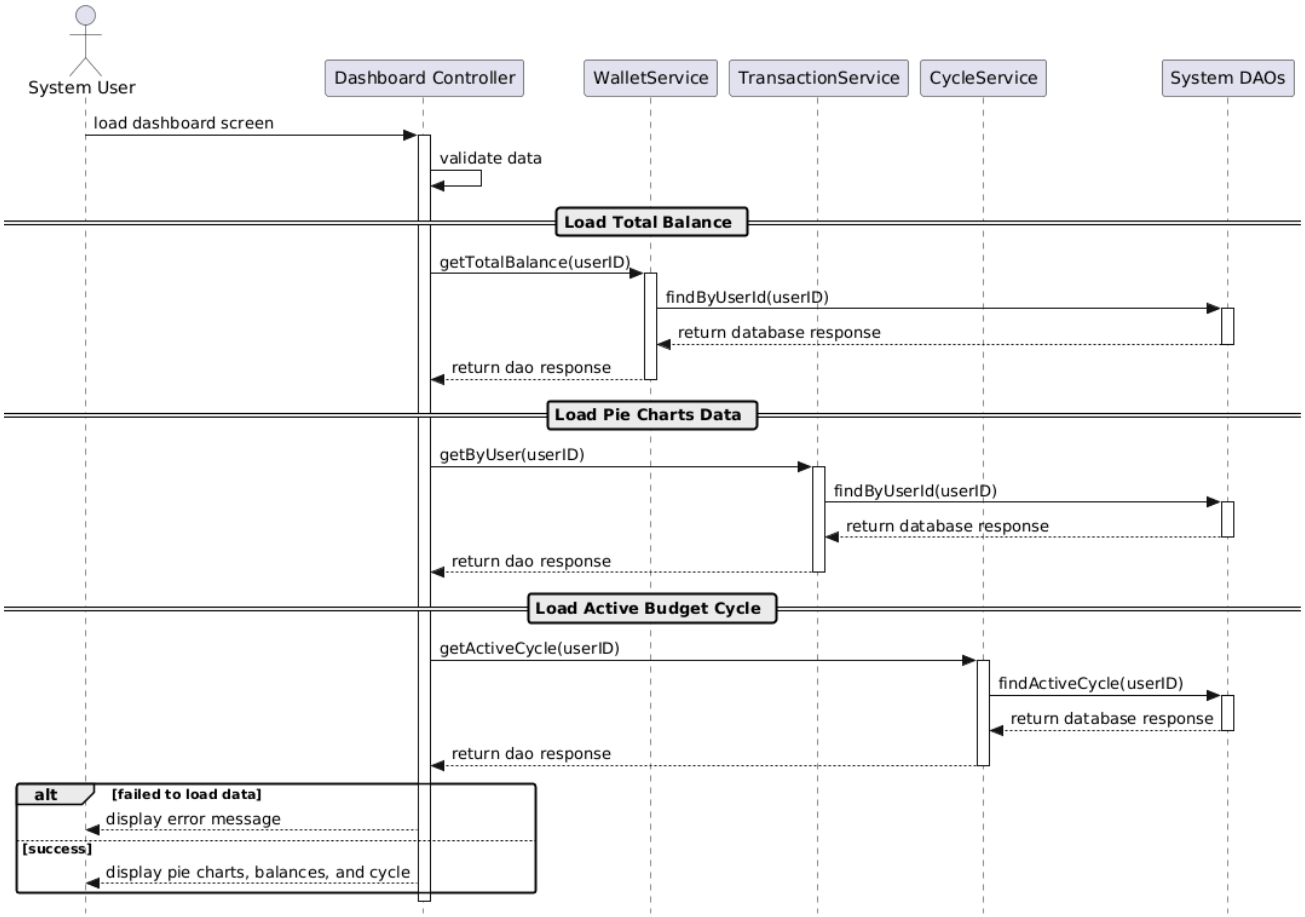
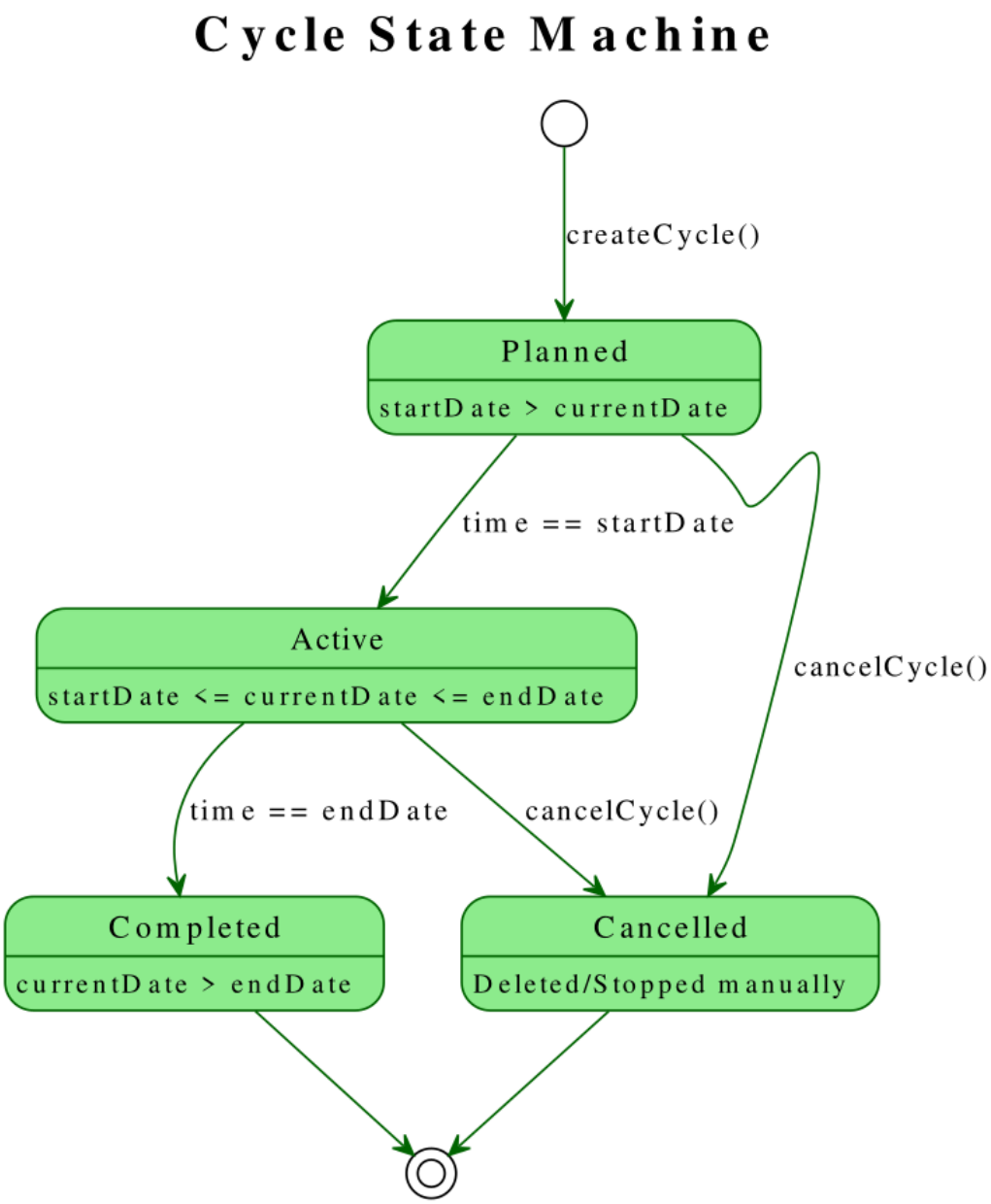
## 9.Dashboard Data Visualization



# Software Design Specification

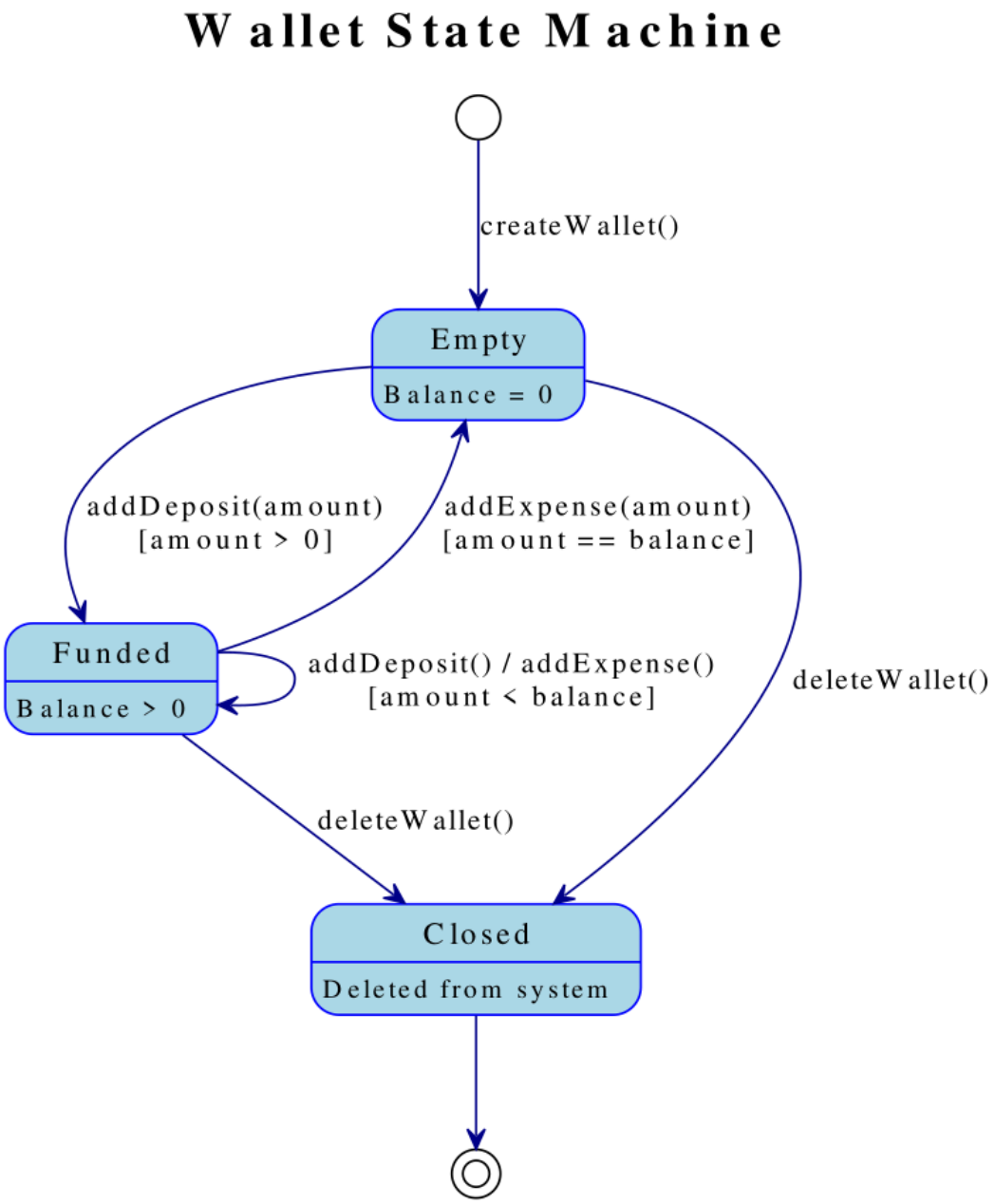
## State Diagram

### 1.Cycle State Diagram



# Software Design Specification

## 2.Wallet State Diagram



# Software Design Specification

## SOLID Principles

**Single Responsibility Principle (SRP)**

- **How it's used: The system strictly isolates responsibilities across its layers. In the Service Layer, each class manages exactly one business domain (e.g., `TransactionService` only handles transaction logic, `CycleService` handles budget math). In the GUI Layer, subclasses of `BaseController` strictly manage UI rendering and user inputs, delegating all actual data processing and business rules back to the services.**

**Open/Closed Principle (OCP)**

- **How it's used: The Database Layer is open for extension but closed for modification through the `IRepository<T>` interface. If you need to add a new data source—such as creating a `MockWalletDAO` for testing or switching from SQLite to PostgreSQL—you simply create a new class that implements `IRepository`. You do not need to modify any of the existing, functioning code in the Service Layer to support the new database.**

**Dependency Inversion Principle (DIP)**

- **How it's used: High-level modules (the Services) do not depend on low-level modules (the concrete DAOs). Instead, Service constructors require the abstraction `IRepository<T>`. For example, `WalletService` doesn't know what `WalletDAO` is; it only knows it has been given a repository that can perform standard database actions. This fully decouples the business logic from the database implementation.**

# Software Design Specification

## Design Patterns

**Singleton Pattern**

- **How it's used:** This pattern ensures that a class has only one globally accessible instance. Spenderman uses it for three critical infrastructure managers to prevent resource conflicts:
  - `DatabaseConnection`: Ensures the app doesn't open hundreds of simultaneous connections to the database, which could cause a crash or lock.
  - `SceneManager`: Coordinates UI routing centrally so multiple screens aren't competing for control of the main application window.
  - `AppEventBus`: Serves as the single source of truth for broadcasting system events.

**DAO (Data Access Object) Pattern**

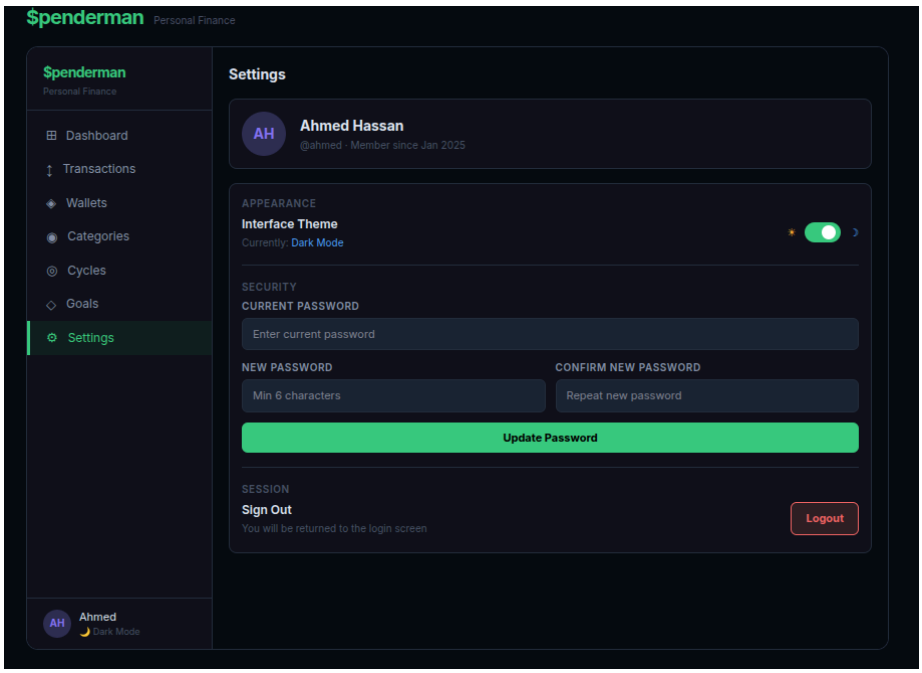
- **How it's used:** The DAO pattern abstracts and encapsulates all access to the data source. Classes like `UserDAO`, `CategoryDAO`, and `TransactionDAO` handle the raw SQL queries and data mapping. This keeps the database logic completely hidden from the Service Layer, making the system much easier to maintain, read, and test.

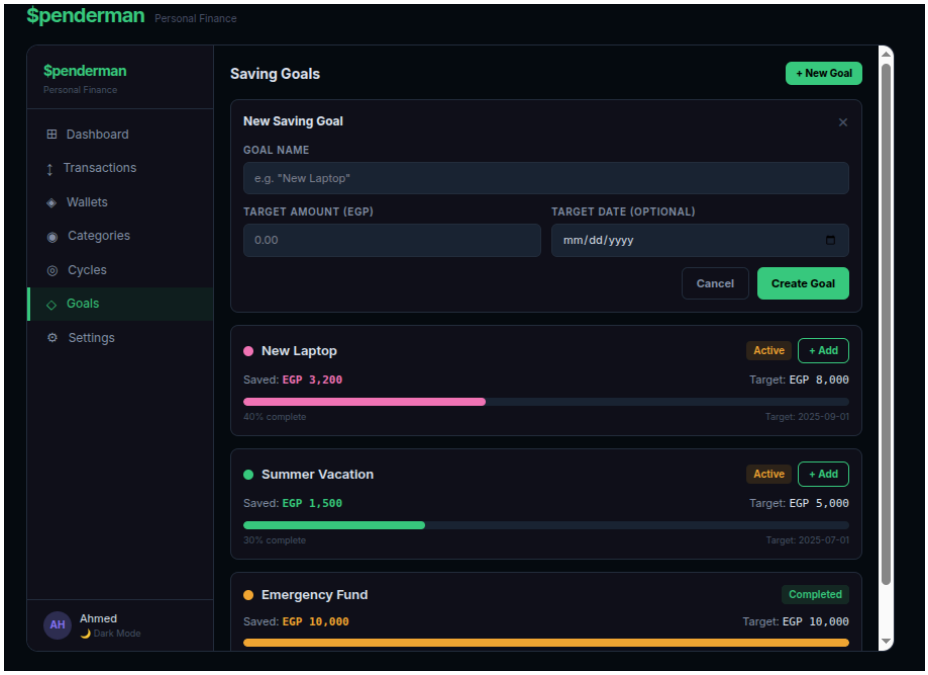
**Observer Pattern**

- **How it's used:** This pattern is used to decouple the backend business logic from the frontend UI through a publish-subscribe system. The `AppEventBus` acts as the publisher (`IObservable`). When a service updates data, it publishes an event (e.g., `"WALLET_UPDATED"`). The GUI Controllers, which implement the `IObserver` interface, listen for these events and automatically trigger a UI refresh (`refreshData()`). Because of this, the backend can update the screen without ever actually touching or knowing about the UI code.

# Software Design Specification

## UI snapshots



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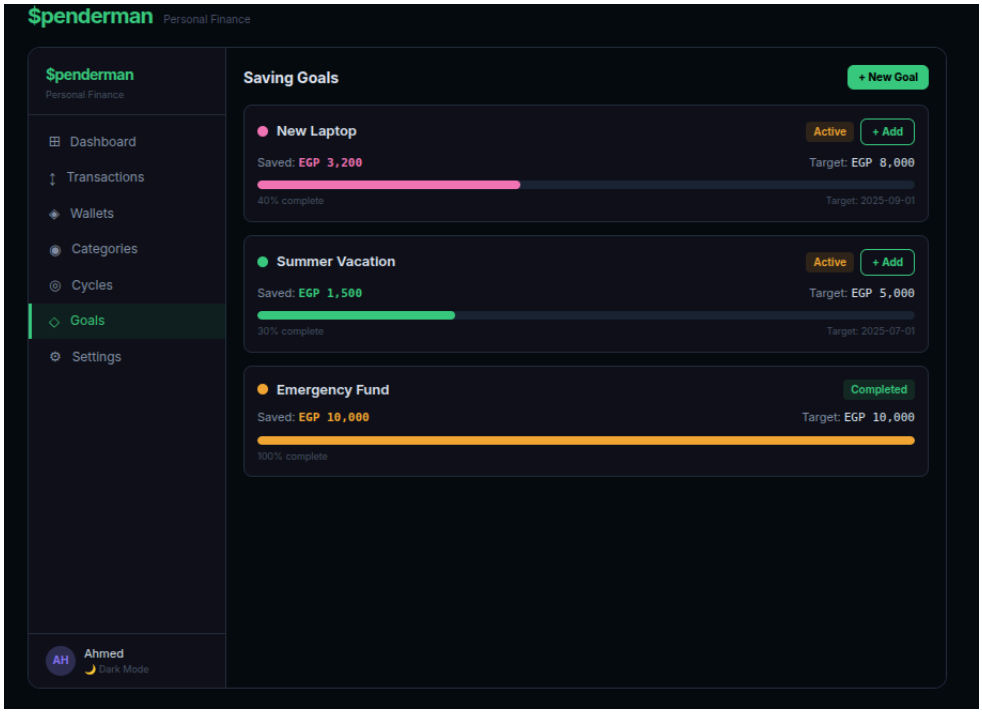


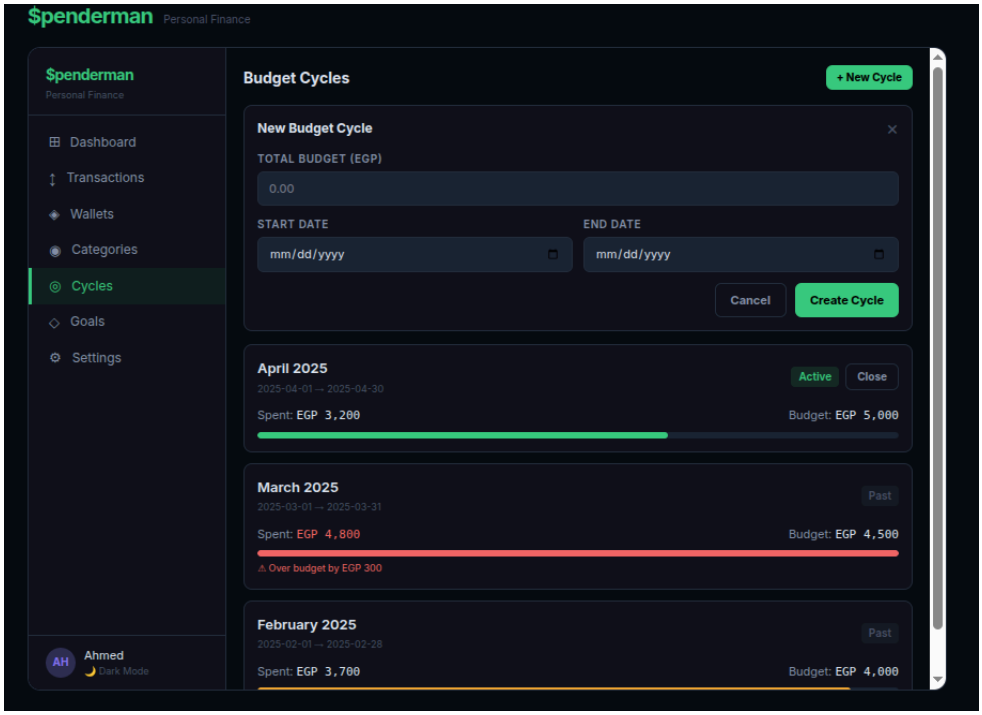

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# Software Design Specification

## Tools

PlantUML , Diagram-io , Figma

## Ownership Report

| Item | ID | Owners |
|---|---|---|
| Ahmed Amr Abdelbadea Ali Shawareb | 20240038 | ER-Diagram , Schema Diagram , UML class Diagram , 3 Sequence Diagram |
| Ahmed Mohamed Salah El-Din | 20240049 | ER-Diagram , UML class Diagram , UI design , 3 sequence Diagram |
| Mohamed Ayser Mohamed El-Hosiny | 20240476 | ER-Diagram , 2 State Diagrams , 3 Sequence Diagram , UML class Diagram |